Emir Sakarya

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Education

Tofaş Fen Lisesi Bursa, Türkiye

Science High School, GPA: 3.98/4.00 *2024 – 2028*

Top 0.18% in Türkiye, LGS 2024 (High School Entrance Exam): scored 492.17/500 among 992,906 candidates

Experience

Founder & Head of Software and Electronics Department 2024 – Present

Stratos UAV (stratosiha.com) *Bursa, Türkiye*

- Founded Stratos UAV from scratch and lead an engineering organization spanning avionics, flight software, and operations
- Architect the complete avionics and autonomy stack for the team's autonomous multirotor (Pixhawk 6C, Raspberry Pi 5, ArduPilot, OpenCV), delivering 19.76-minute hover endurance, 12 km autonomous range, and precision landing
- Direct development of a 170 km/h FPV racing drone with 8 ms video latency for the MEB Robot competition
- Took the program to the TEKNOFEST 2026 rotorcraft final, placing 16th of 3,715 teams, and to the NASA Space Apps Turkey final in its first year of operation
- Raised over 400,000 TL in corporate sponsorship and grew the founding team from 3 to 60+ members, with a core of about 30 engineers
- Featured by Anadolu Ajansı and Bursa Hakimiyet for public autonomous flight demonstrations

Team Captain 2024 – Present

TEKNOFEST *Turkey*

- Lead competition teams across five TEKNOFEST categories: Technology for Humanity (2025), High School UAV (2025 and 2026), E-Commerce (2026), and Technological Applications in the Fight Against Addiction (2026)

Technical Director 2024 – 2026

Turbine Engine Technologies Club *Bursa, Türkiye*

- Built and maintained the official team website with Next.js and Tailwind CSS for team and project showcase
- Wrote Python simulations of jet engine performance and turbine behavior, improving predicted thrust accuracy by 27% over the club's prior model
- Developed an FPV UAV with real-time video transmission for the MEB Robot competition
- Designed and built a competition-level VEX V5 robot with custom mechanical design and sensor integration, placing 2nd nationally

Collaborator 2026 – Present

Team1 Avalanche

- Contribute as a developer to Team1 Avalanche, working on Avalanche ecosystem projects

Founding Technical Developer Member 2025 – Present

Viyaro (Autonomous Mobility Startup) *Remote*

- Founding technical member of Viyaro, building autonomous shuttle systems for urban transport, working fully remotely with the team
- Selected and fully funded by BOSİAD (Bursa Organize Sanayi Bölgesi Sanayicileri ve İş İnsanları Derneği)
- Now in daily production use across 10+ companies, logging 50,000+ rides to date

Projects

Tofaş Fen Webapp | *React 19, Express, MongoDB, Kubernetes, GraphQL* | tofasfen.com 2024 – Present

- Built a production school management system with 22 MongoDB models and 6 user roles, serving over 2,000 active users at 99.95% uptime
- Implemented JWT authentication with 2FA, a GraphQL API, and WebSocket real-time features
- Deployed on Kubernetes for scalable, containerized operation, handling 40,000+ requests per day

TableTab | *Next.js 16, React 19, Drizzle ORM, Neon Postgres, TypeScript* 2026

- Built a QR-code table-ordering and bill-splitting web app for restaurants: the customer scans the table QR, browses a localized menu, orders, and splits the check by item, evenly, or in full
- Designed a 12-table Postgres schema (Drizzle ORM on Neon) covering table sessions, orders, split-payment item locks, and waiter calls, with role-based panels for admin, waiter, and kitchen
- Streamed live order status to the customer over server-sent events on the Node runtime; wired card payments through iyzico alongside fast-IBAN and cash flows
- Shipped full Turkish/English i18n via a locale cookie (no URL change, so printed QR links stay valid), allergen/diet filtering over 7 fixed tags, JWT auth with bcrypt, and a daily Vercel cron that flags unpaid tables

Local AI Assistant | *Python, FastAPI, Ollama* 2025

- Built a WhatsApp-based assistant running locally on an RTX 3060 Ti, cutting average response latency to under 800ms
- Implemented task management, calendar, habit tracking, expense tracking, and a mood journal, used daily for over 6 months

Zero-G Pharma | *Python, VPython* 2025

- Built a 3D simulator of space drug crystal growth in microgravity versus Earth gravity
- Placed First in Turkey at the NASA Space Apps Challenge 2025, advancing to the global top 50 out of 6,000+ teams

Teluvane | *FastAPI, LangGraph, Claude, Postgres, Next.js, Supabase, Web3* | teluvane.com 2026 – Present

- Founded and built Teluvane, a multi-tenant compliance auditing platform for AI agents: every LLM call, tool call, and tool result is SHA-256 hash-chained per session so tampering is detectable, not just logged
- Built a policy tribunal that audits sessions against EU AI Act, ISO 42001, NIST AI RMF, and SOC 2 policy packs, producing cited verdicts from a deterministic offline detector or a live LangGraph + Claude pipeline
- Shipped the product end to end: Supabase-authenticated orgs, per-org API keys, LemonSqueezy subscription billing, and a Postgres store with org-scoping enforced at the code level, backed by 117 passing automated tests
- Built an MCP server and Python/JS SDKs for zero-code agent integration, plus optional on-chain session anchoring on Avalanche with a public, trustless verification page

DurAn | *React Native, TypeScript, ESP32S3, Bluetooth LE* | dur-an.vercel.app 2026 – Present

- TEKNOFEST Finalist, Bağımlılıklarla Mücadelede Teknolojik Uygulamalar Yarışması, as team captain of Stratos Root
- Build a two-part early-warning system for gambling relapse: an ESP32S3 wristband (MAX30102 PPG, GSR, MPU6050, DS18B20) fused on-device with a React Native app's behavioral signal to detect craving spikes inside a 3-minute window
- Run a fully on-device fusion engine: no server, no stored raw data, only event timestamps persisted; 14 passing unit tests across the physiological and behavioral pipelines
- Hit 94% detection accuracy on simulated relapse events (no clinical data yet), behind a three-tier escalating intervention flow (nudge, pause screen, breathing exercise + YEDAM hotline)
- Researched prior art (including an existing gambling-detection wearable patent) to differentiate DurAn's self-monitoring, privacy-first approach from surveillance-based alternatives

Awards & Honors

1st in Turkey, NASA Space Apps Challenge 2025 (global top 50 of 6,000+ teams)

2x TEKNOFEST Finalist (Rotorcraft, 16th of 3,715 teams; Technology Against Addiction)

2nd in Turkey, VEX V5 Robotics National Championship

Certifications

CS50P: Introduction to Programming with Python – Harvard University

AI Fluency: Framework & Foundations – Anthropic

Claude Code 101 – Anthropic

Galactic Problem Solver – NASA Space Apps Challenge 2025

Interstellar Problem Solver – TUA Astro Hackathon, T.C. Sanayi ve Teknoloji Bakanlığı

Seed Sprint Builder Residency – OKX TR & Team1 Türkiye

Applied Entrepreneurship Training (3 hours) – T3 Girişim Merkezi, TEKNOFEST 2026 Şanlıurfa

Participant – GençTek Ankara 2025

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript
Web & Mobile: React, React Native, Next.js, Node.js, Express, FastAPI, GraphQL, PostgreSQL, MongoDB, Redis
Embedded & Hardware: Embedded C/C++, ArduPilot, CAN bus, KiCad PCB design, soldering & rework
Blockchain / Web3: Avalanche, on-chain verification, Web3 integration
Infra: Docker, Kubernetes, Linux, Git, GitHub Actions, Vercel **Spoken**: Turkish (native), English (B2), German (A2)